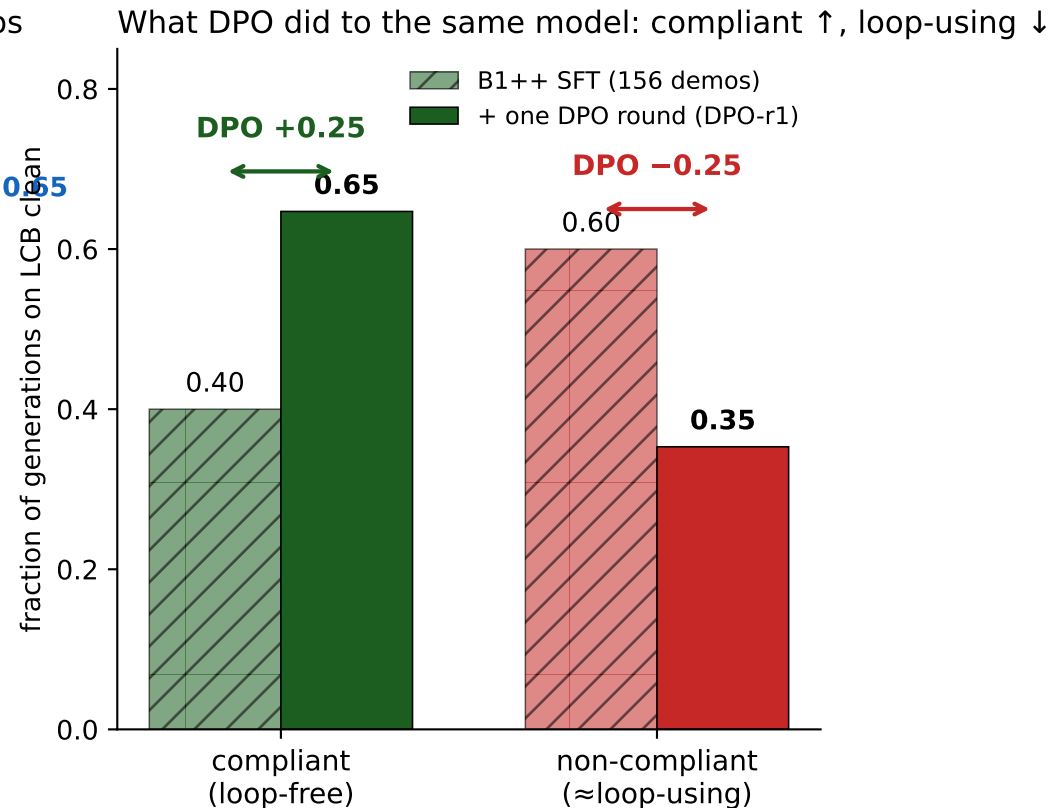
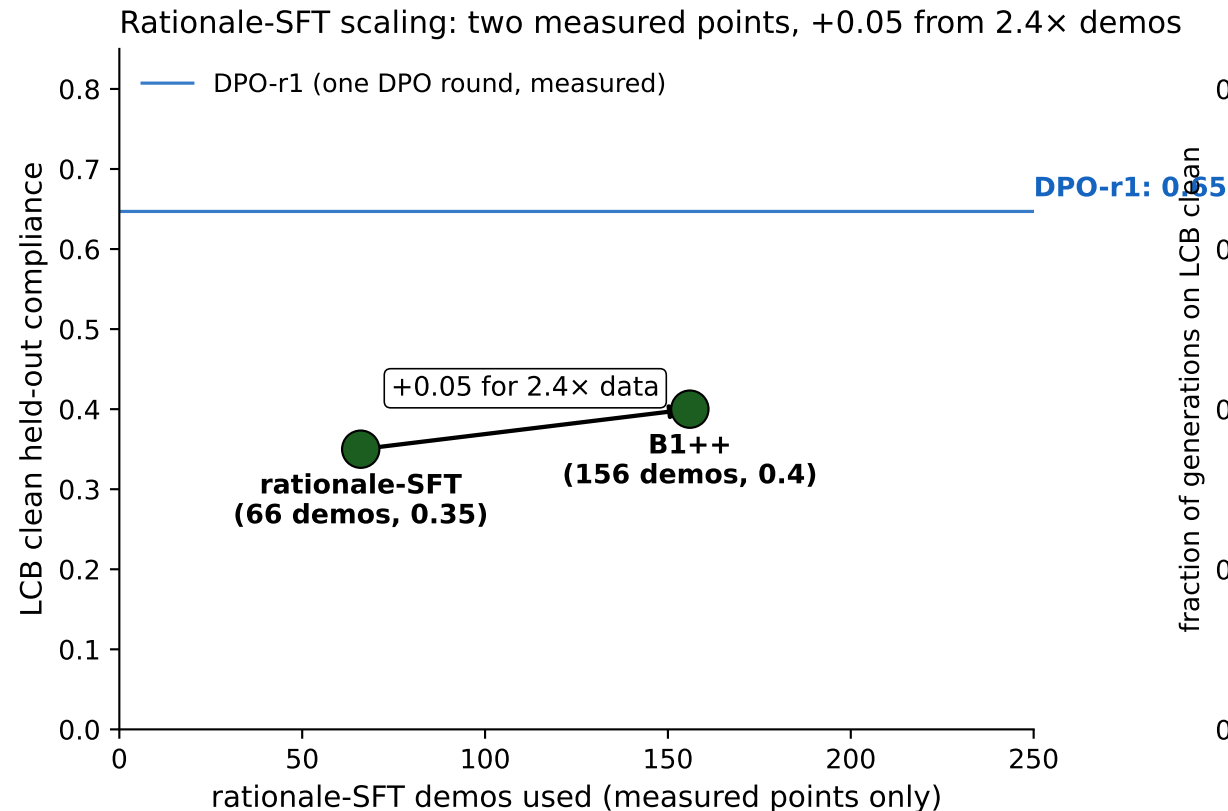


Why more rationale-SFT won't close the gap to DPO-r1



Left: two measured rationale-SFT scaling points (66, 0.35) and (156, 0.40) on LCB clean held-outs. The dashed DPO-r1 line (0.65) is what one DPO round on the same B1++ adapter achieves. We do not extrapolate the SFT curve — the only honest claim from two points is the +0.05/2.4× delta.

Right: same B1++ adapter before vs after one DPO round (AST-rechecked rates on the 17 clean held-out problems, $n=8$ samples each). DPO's loss has a direct negative term on rejected (loop-using) tokens; SFT's loss can only push compliant tokens up. That structural difference is the gap data can't fill.